

Digital Signal Processing

Spectral Estimation Techniques II

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Periodogram

MS spectrum $S_{ms}[k]$ unit is Watts

We want the density in watts per hertz

$$\sum \hat{S}_{ms}[k] = \sum \hat{S}[k] \cdot \Delta f$$

W

W/hz

hz

Since the periodogram does not learn parameters, it is a non-parametric estimator

$$\hat{S}_{ms}[k] = \left| \frac{FFT\{x_N[n]\}}{N} \right|^2$$

Frequency bin width: $\Delta f = \frac{f_s}{n_{perseg}}$

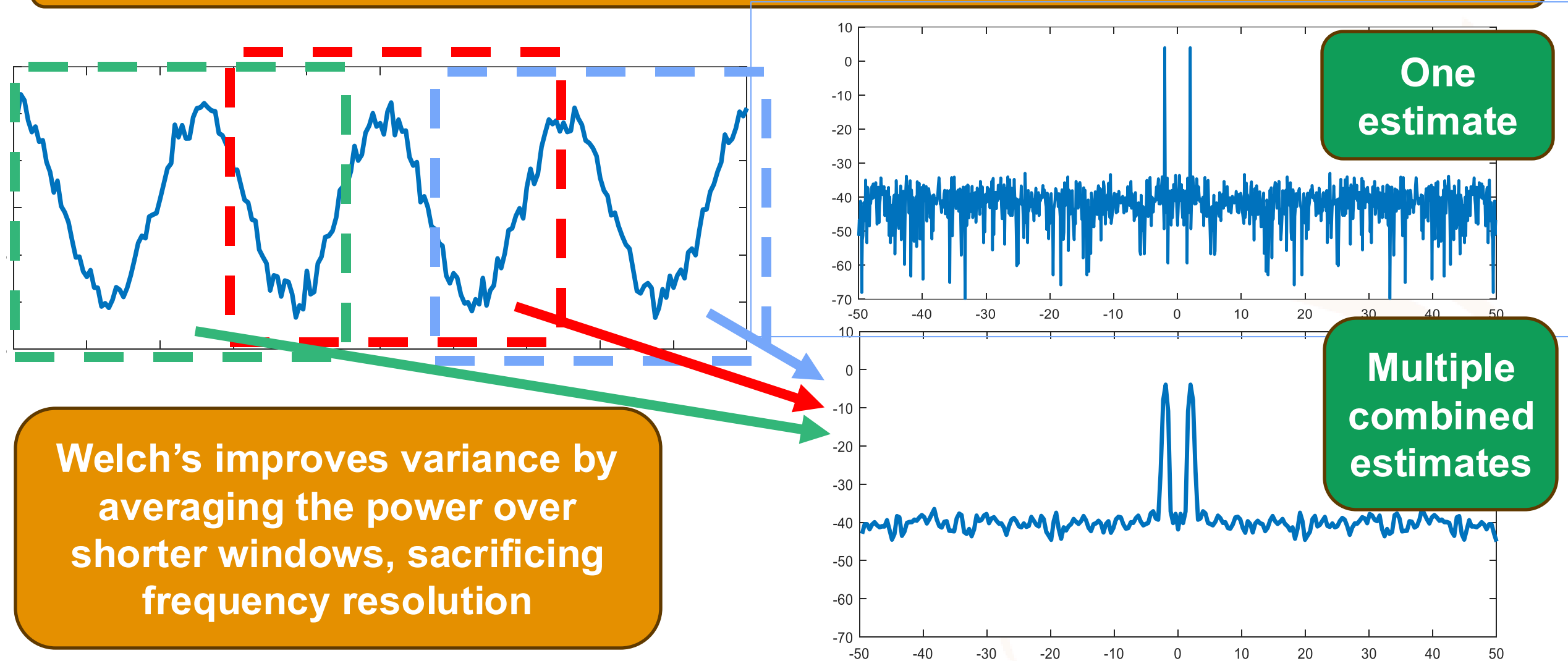
$$\hat{S}(f) \approx \hat{S}[k] = \frac{\hat{S}_{ms}[k]}{\Delta f} = \frac{N}{F_s} \hat{S}_{ms}[k]$$

Periodogram definition

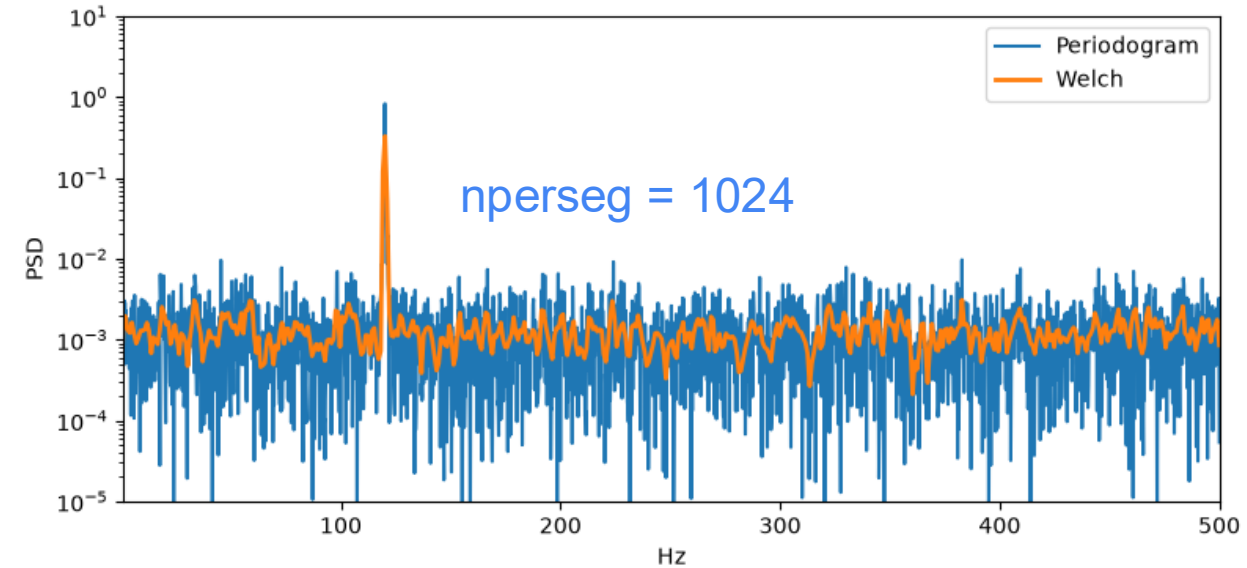
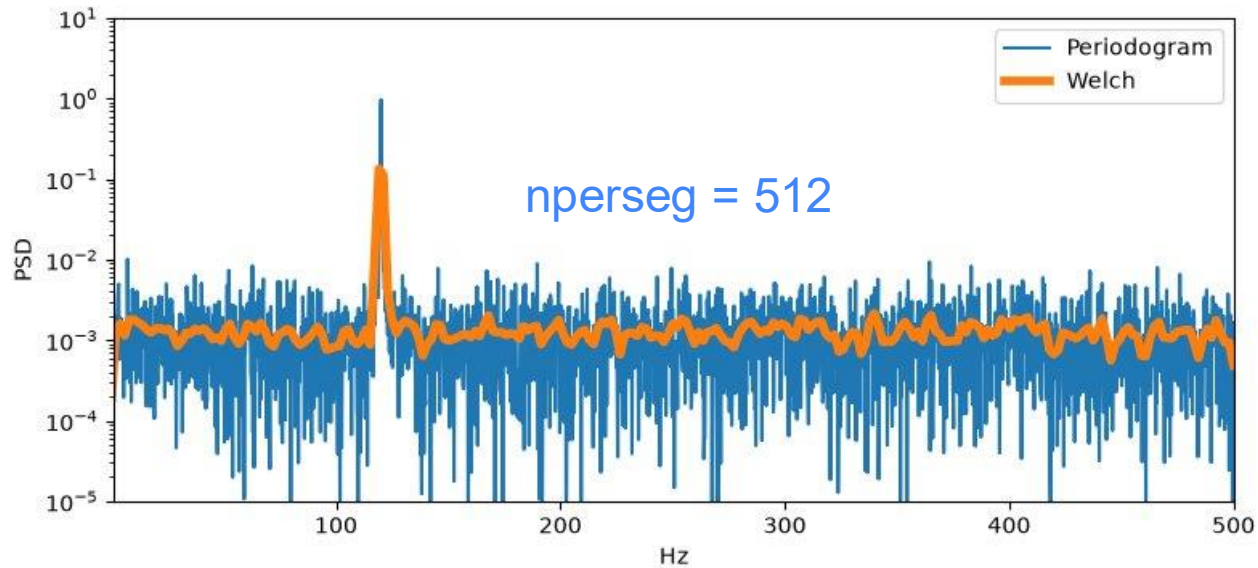
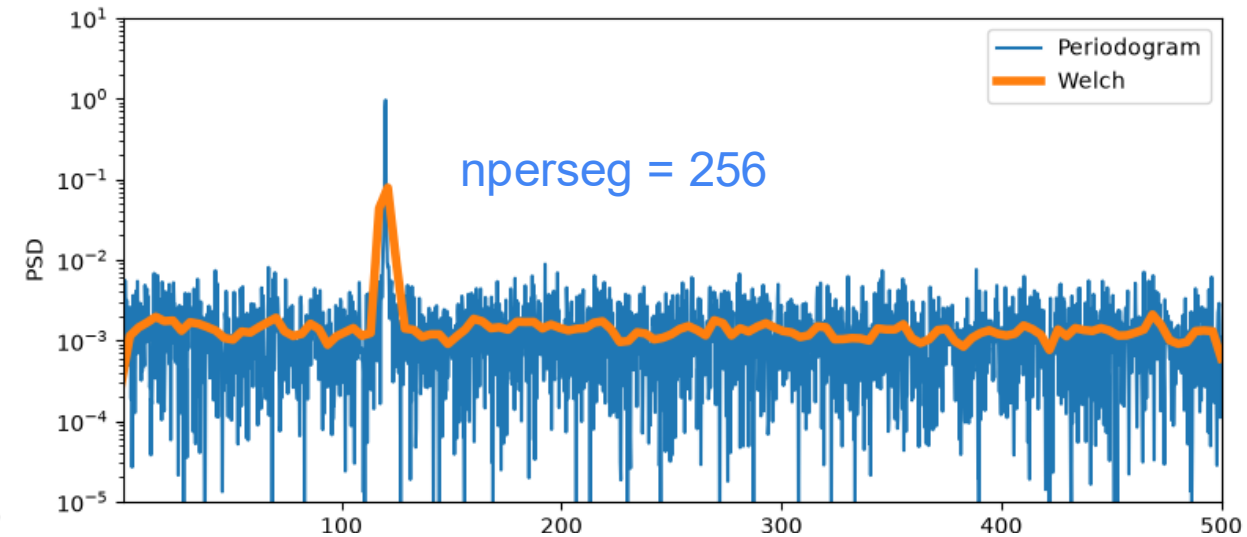
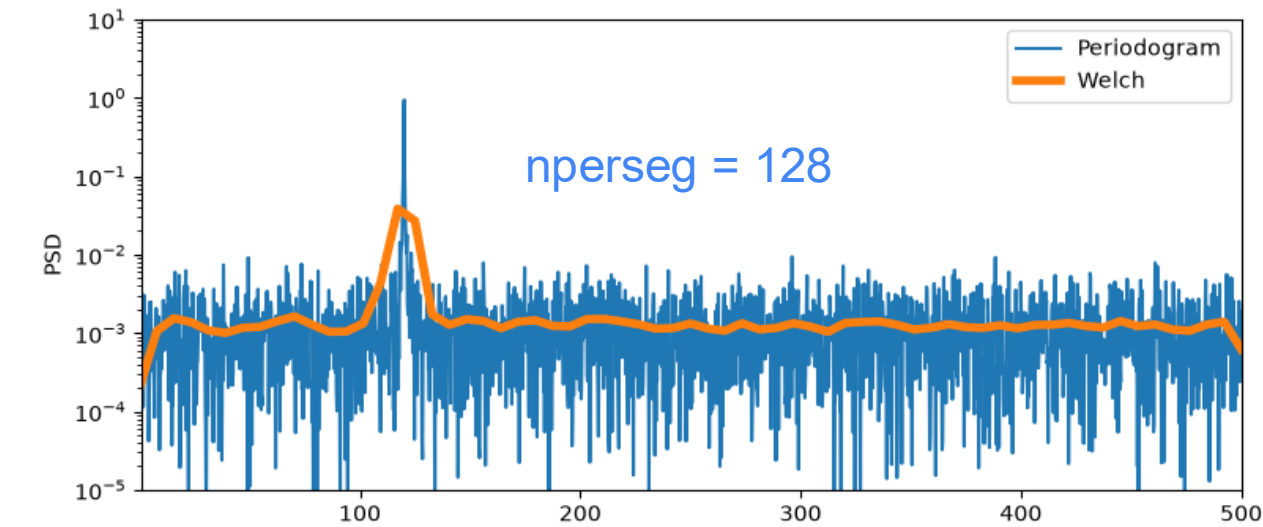
$$\hat{S}(f) \approx \hat{S}[k] = \frac{1}{N BW} |FFT\{x_N[n]\}|^2$$

Welch: averages of periodograms

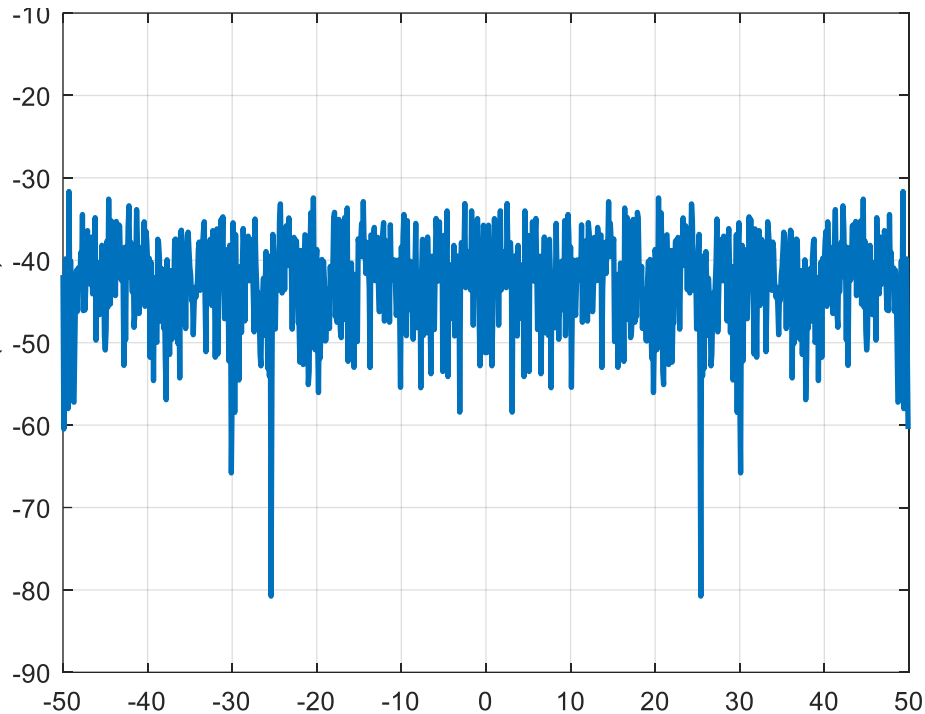
One periodogram is one estimate/measurement of $S(f)$ of a random process



$$\text{Frequency bin width: } \Delta f = \frac{f_s}{\text{nperseg}}$$

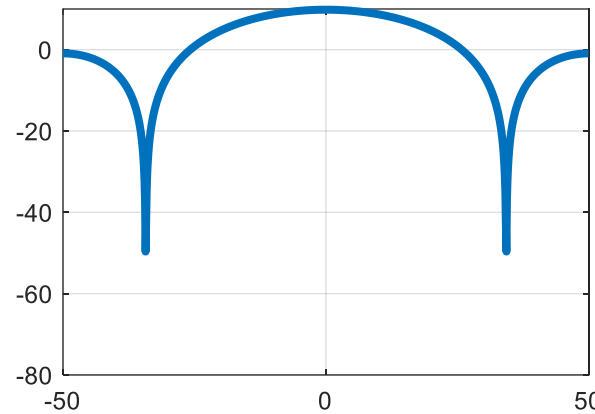


Filtering white noise shapes the noise spectrum like the filter

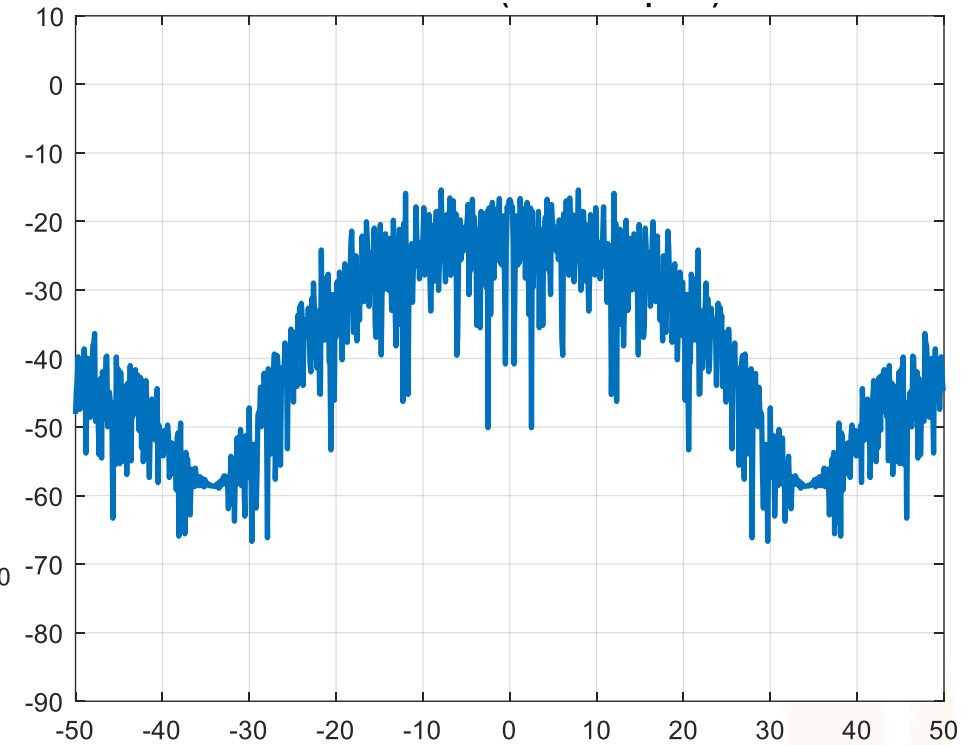


$$\hat{S}_x(f)$$

White noise



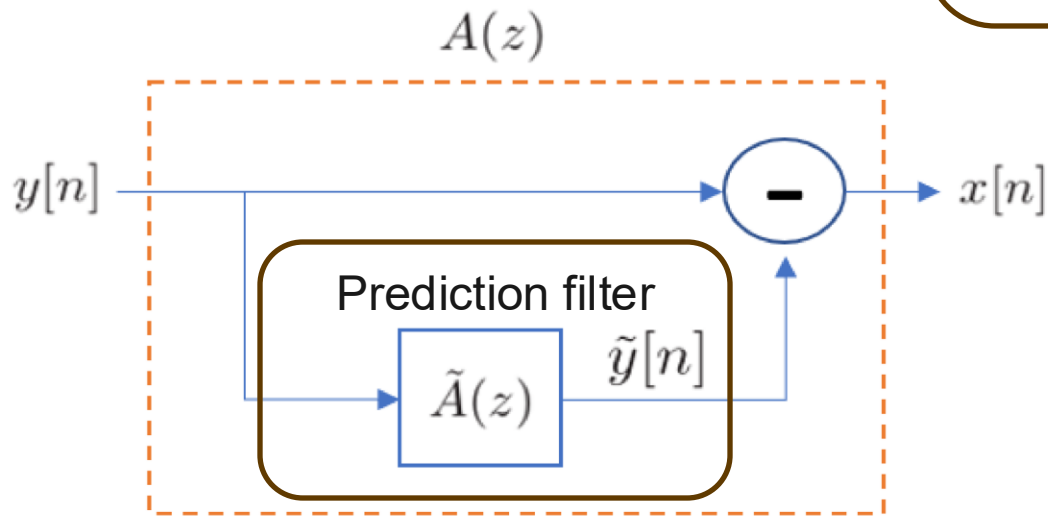
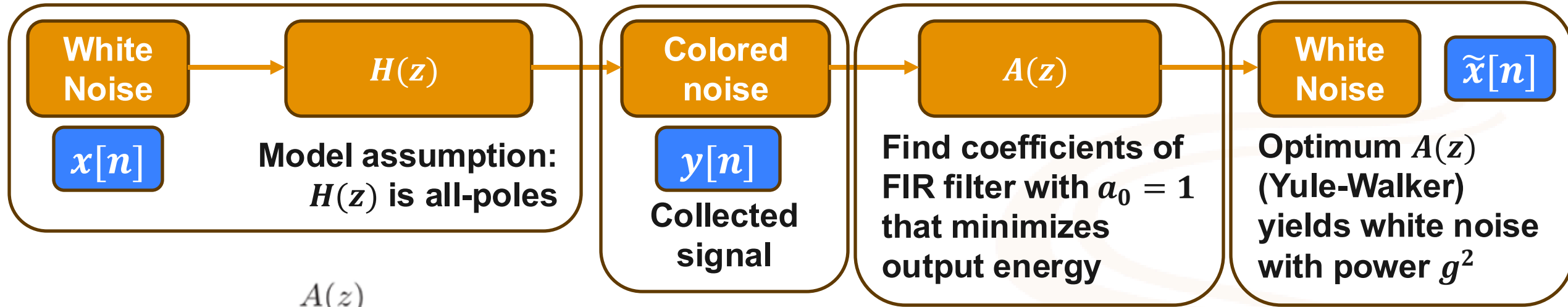
$$|H(f)|^2$$



$$\hat{S}_y(f) = |H(f)|^2 \hat{S}_x(f)$$

Colored noise

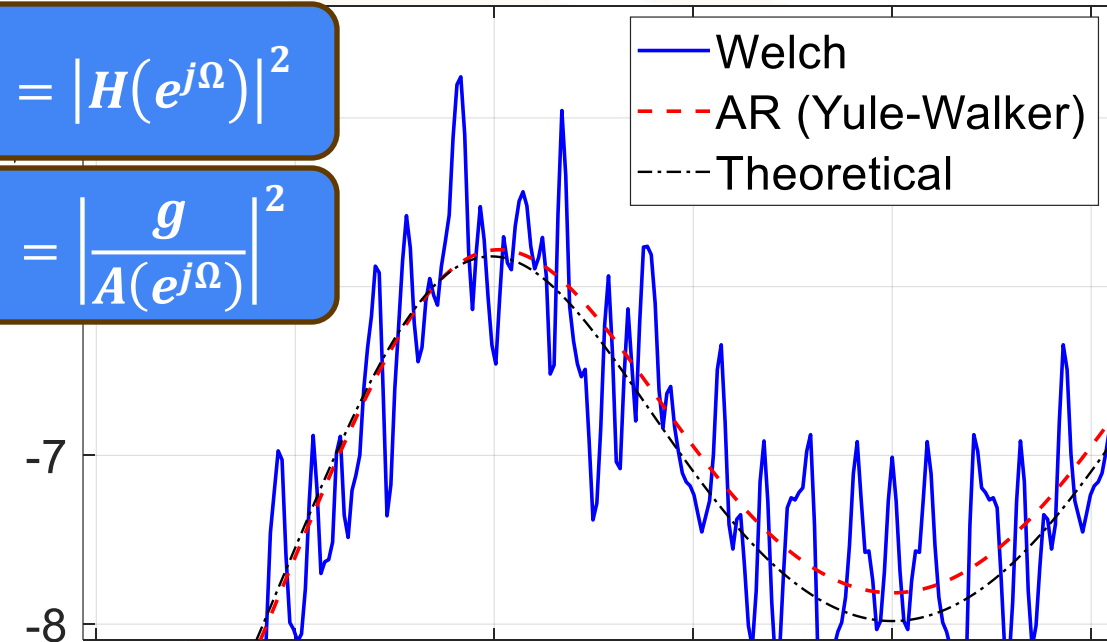
Autoregressive (AR) Modeling



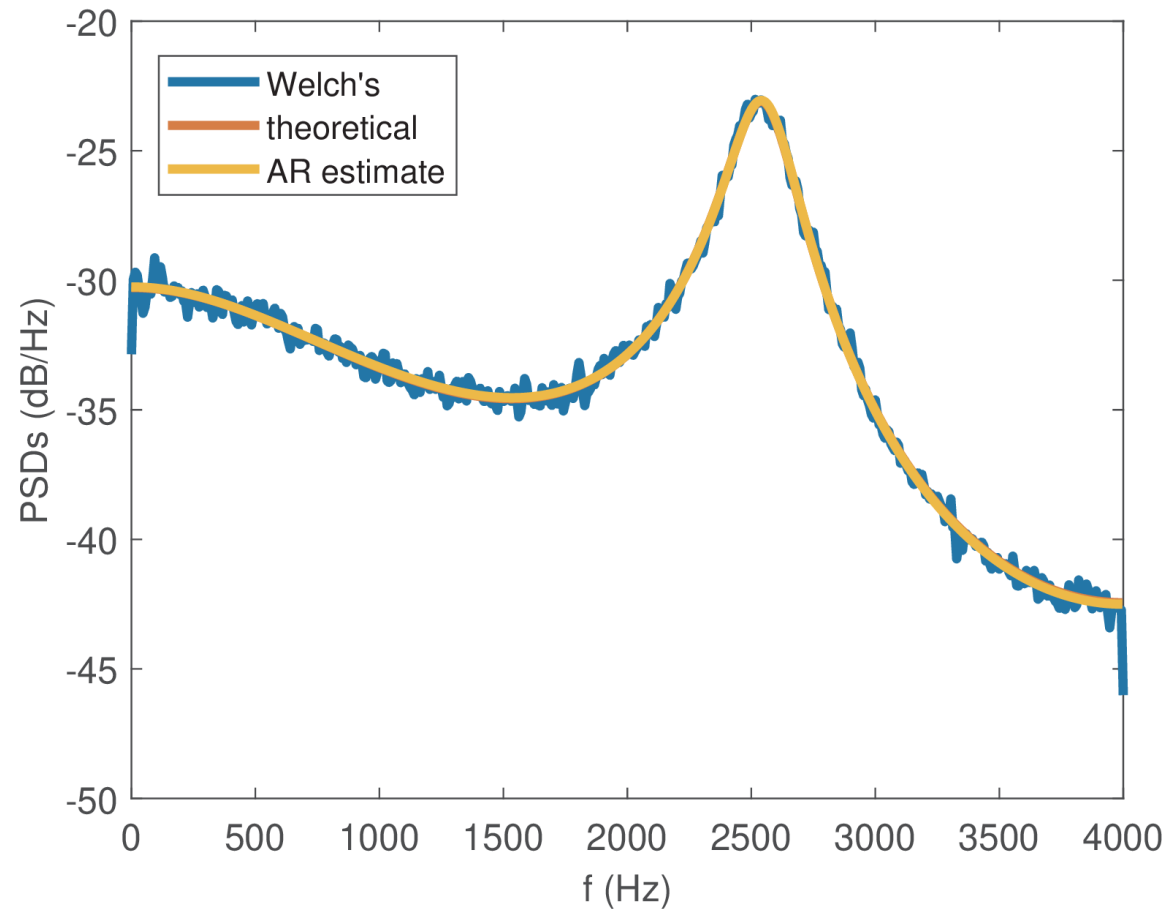
$$S(e^{j\Omega}) = |H(e^{j\Omega})|^2$$

$$\hat{S}(e^{j\Omega}) = \left| \frac{g}{A(e^{j\Omega})} \right|^2$$

AR model is widely used in speech coding to mimic voice spectrum

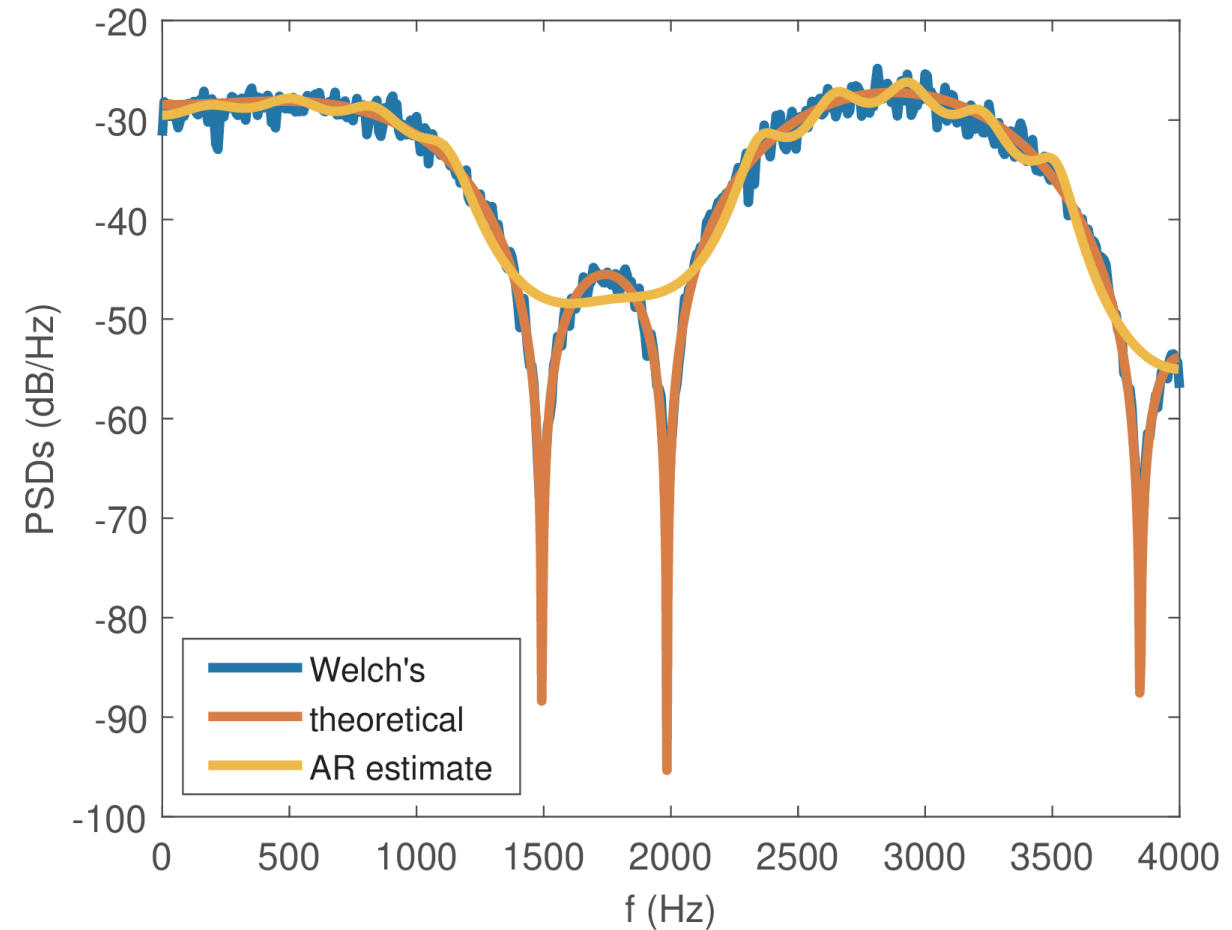


Satisfying the assumptions of the AR model



Far fewer coefficients compared to Welch's method

Not satisfying the assumptions of the AR model (system has zeros)

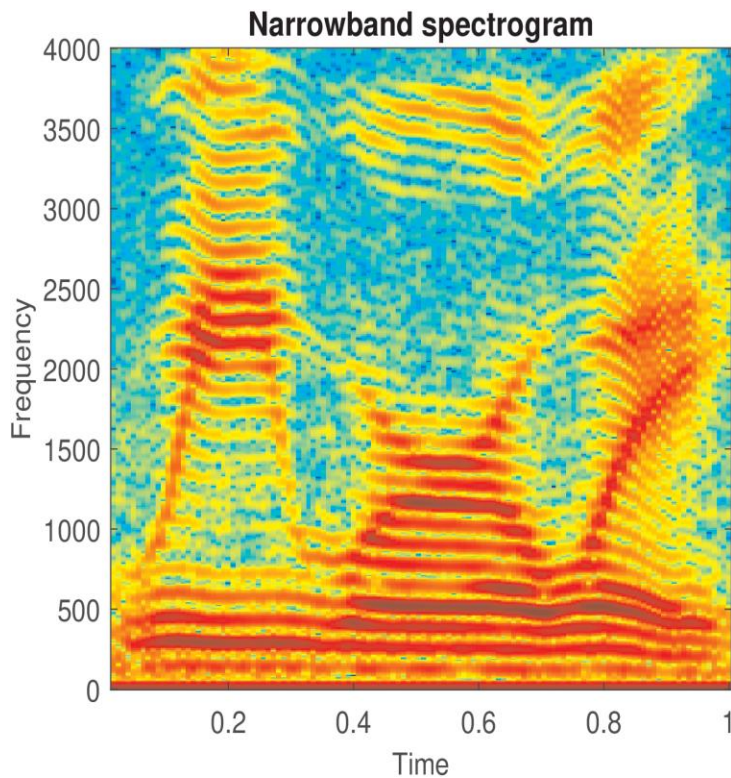


Poles cannot create zeros, so AR model performs poorly near notches

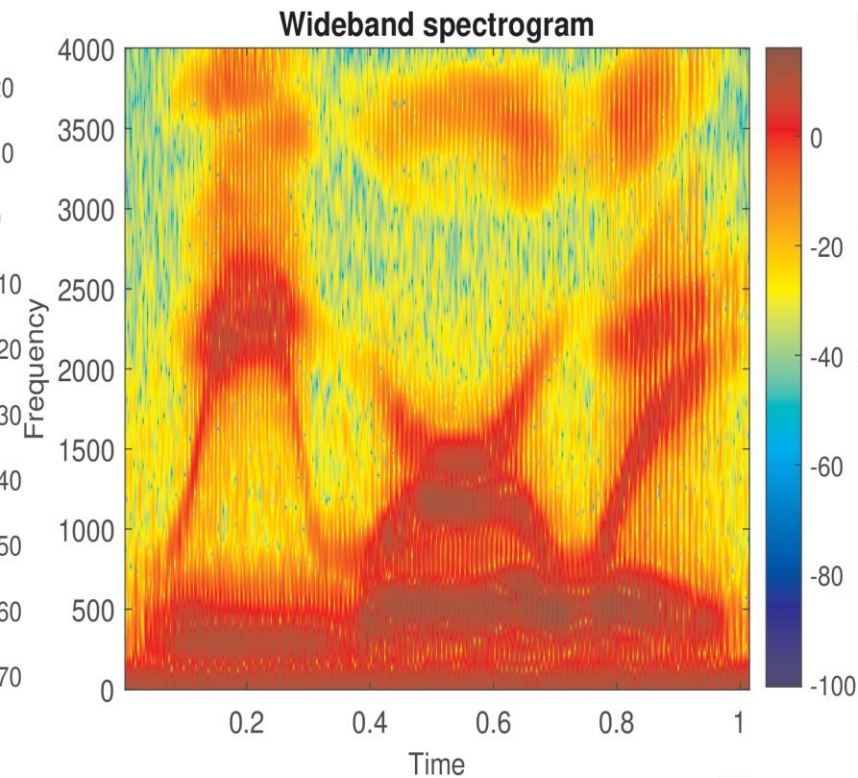
Spectrogram

Nonstationary processes have time-varying spectrum

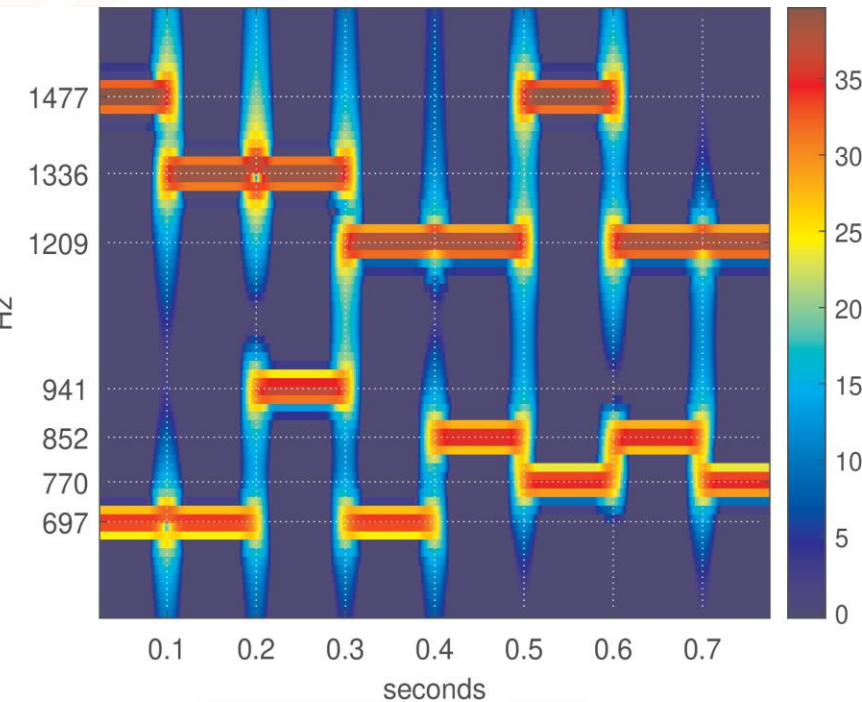
Spectrograms take multiple windows but do not combine them like Welch's method



Speech



Speech



Piano



Thank you!

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